Heatmaps_GDS_Final

2026-03-14

Purpose

This notebook creates three LSOA-level heat maps for the HMP Fosse Way study area:

1. Average house price by LSOA **before** the chosen event date
2. Average house price by LSOA **after** the chosen event date
3. Relative house price inflation by LSOA (**after vs before**)

```
library(data.table)
library(sf)
library(ggplot2)
library(dplyr)
library(scales)

knitr::opts_chunk$set(
  echo = TRUE,
  fig.width = 10,
  fig.height = 8,
  fig.align = "center"
)
```

File paths and parameters

```
base_path <- "/Users/rohankale/Documents/BSE/Geospatial DS/Final Project"

ppd_file <- file.path(base_path, "matched_ppd.csv")
distance_file <- file.path(base_path, "property_prison_distance.csv")
lsoa_file <- file.path(base_path, "fosse_way_lsoa_bbox_candidates.geojson")

event_date <- as.Date("2023-05-01")
min_sales_per_lsoa <- 3
max_distance_m <- 11000
```

Check files exist

```
file.exists(ppd_file)
```

```
## [1] TRUE
```

```r
file.exists(distance_file)
```

```
## [1] TRUE
```

```r
file.exists(lsoa_file)
```

```
## [1] TRUE
```

Load data

```r
ppd <- fread(ppd_file)
distance_dt <- fread(distance_file)
lsoa_sf <- st_read(lsoa_file, quiet = TRUE)
```

Inspect columns

```r
colnames(ppd)
```

```
##  [1] "unique_id"            "price_paid"           "deed_date"
##  [4] "postcode"             "property_type"        "new_build"
##  [7] "estate_type"          "saon"                 "paon"
## [10] "street"               "locality"             "town"
## [13] "district"             "county"               "transaction_category"
## [16] "linked_data_uri"      "lsoa21cd"
```

```r
colnames(distance_dt)
```

```
##  [1] "match_key"                     "postcode"
##  [3] "unique_id"                     "LMK_KEY"
##  [5] "geocode_query"                 "latitude"
##  [7] "longitude"                     "display_name"
##  [9] "geocode_status"                "distance_to_prison_m"
## [11] "postcode_distance_to_prison_m" "postcode_latitude"
## [13] "postcode_longitude"            "postcode_display_name"
## [15] "postcode_geocode_status"       "keep_for_full_geocode"
```

```r
colnames(lsoa_sf)
```

```
##  [1] "FID"       "LSOA21CD"  "LSOA21NM"  "LSOA21NMW" "BNG_E"     "BNG_N"
##  [7] "LAT"       "LONG"      "GlobalID"  "geometry"
```

Merge house-price and distance data

```r
houses <- merge(
  ppd,
  distance_dt,
  by = "unique_id",
  all = FALSE
)
```

Check merged columns

```r
colnames(houses)
```

```
##  [1] "unique_id"                     "price_paid"
##  [3] "deed_date"                     "postcode.x"
##  [5] "property_type"                 "new_build"
##  [7] "estate_type"                   "saon"
##  [9] "paon"                          "street"
## [11] "locality"                      "town"
## [13] "district"                      "county"
## [15] "transaction_category"          "linked_data_uri"
## [17] "lsoa21cd"                      "match_key"
## [19] "postcode.y"                    "LMK_KEY"
## [21] "geocode_query"                 "latitude"
## [23] "longitude"                     "display_name"
## [25] "geocode_status"                "distance_to_prison_m"
## [27] "postcode_distance_to_prison_m" "postcode_latitude"
## [29] "postcode_longitude"            "postcode_display_name"
## [31] "postcode_geocode_status"       "keep_for_full_geocode"
```

Standardize the LSOA code field in the shapefile

```r
possible_lsoa_cols <- c("lsoa21cd", "LSOA21CD", "lsoa_code", "LSOA_CODE")
matched_lsoa_col <- intersect(possible_lsoa_cols, names(lsoa_sf))

if (length(matched_lsoa_col) == 0) {
  stop("No LSOA code column found in the shapefile.")
}

if (matched_lsoa_col[1] != "lsoa21cd") {
  lsoa_sf <- lsoa_sf %>%
    rename(lsoa21cd = all_of(matched_lsoa_col[1]))
}

lsoa_sf <- st_make_valid(lsoa_sf)
```

Standardize key columns in the merged table

```r
possible_date_cols <- c("deed_date", "DEED_DATE", "date_of_transfer", "Date of Transfer")
matched_date_col <- intersect(possible_date_cols, names(houses))

if (length(matched_date_col) == 0) {
  stop("No deed-date column found after merging ppd and distance data.")
}

if (matched_date_col[1] != "deed_date") {
  setnames(houses, matched_date_col[1], "deed_date")
}

possible_price_cols <- c("price_paid", "PRICE_PAID", "sold_amount")
matched_price_col <- intersect(possible_price_cols, names(houses))

if (length(matched_price_col) == 0) {
  stop("No house-price column found after merging ppd and distance data.")
}

if (matched_price_col[1] != "price_paid") {
  setnames(houses, matched_price_col[1], "price_paid")
}

possible_lsoa_house_cols <- c("lsoa21cd", "LSOA21CD")
matched_lsoa_house_col <- intersect(possible_lsoa_house_cols, names(houses))

if (length(matched_lsoa_house_col) == 0) {
  stop("No lsoa21cd column found after merging ppd and distance data.")
}

if (matched_lsoa_house_col[1] != "lsoa21cd") {
  setnames(houses, matched_lsoa_house_col[1], "lsoa21cd")
}
```

Basic cleaning

```r
houses[, deed_date := as.Date(deed_date)]
houses[, price_paid := as.numeric(price_paid)]

required_cols <- c("lsoa21cd", "price_paid", "deed_date")
missing_required <- setdiff(required_cols, names(houses))

if (length(missing_required) > 0) {
  stop(
    paste(
      "Missing required column(s):",
      paste(missing_required, collapse = ", ")
    )
  )
}

if ("distance_to_prison_m" %in% names(houses)) {
```

```r
  houses <- houses[
    !is.na(distance_to_prison_m) & distance_to_prison_m <= max_distance_m
  ]
}

if ("geocode_status" %in% names(houses)) {
  houses <- houses[geocode_status == "ok"]
}

houses <- houses[
  !is.na(lsoa21cd) &
  !is.na(price_paid) &
  !is.na(deed_date)
]

houses <- houses[lsoa21cd %in% lsoa_sf$lsoa21cd]

houses[, period := ifelse(deed_date < event_date, "Before", "After")]
houses[, deed_year := format(deed_date, "%Y")]
```

Quick summary

```r
summary_dt <- houses[, .(
  rows = .N,
  unique_sales = uniqueN(unique_id),
  unique_lsoas = uniqueN(lsoa21cd),
  min_price = min(price_paid, na.rm = TRUE),
  median_price = median(price_paid, na.rm = TRUE),
  max_price = max(price_paid, na.rm = TRUE)
)]

summary_dt
```

```
##      rows unique_sales unique_lsoas min_price median_price max_price
##     <int>        <int>        <int>     <num>        <num>     <num>
## 1: 135326        94337          347       100       190000   2.9e+07
```

Build average price by LSOA before the event

```r
before_prices <- houses[
  period == "Before",
  .(
    n_before = .N,
    avg_price_before = mean(price_paid, na.rm = TRUE)
  ),
  by = lsoa21cd
]

before_prices <- before_prices[n_before >= min_sales_per_lsoa]
```

```r
before_prices
```

```
##       lsoa21cd n_before avg_price_before
##         <char>    <int>            <num>
##   1: E01025991      287         231480.1
##   2: E01025879      306         177707.1
##   3: E01025640      411         271337.3
##   4: E01025619      700         205453.3
##   5: E01025780      456         259235.9
##  ---
## 341: E01032868        8         237312.5
## 342: E01025787       35         268874.3
## 343: E01013653      349         173812.3
## 344: E01034023        6         194400.0
## 345: E01034022       16         843318.8
```

Build average price by LSOA after the event

```r
after_prices <- houses[
  period == "After",
  .(
    n_after = .N,
    avg_price_after = mean(price_paid, na.rm = TRUE)
  ),
  by = lsoa21cd
]

after_prices <- after_prices[n_after >= min_sales_per_lsoa]

after_prices
```

```
##       lsoa21cd n_after avg_price_after
##         <char>   <int>           <num>
##   1: E01013603      37        195925.5
##   2: E01013694      45        249726.7
##   3: E01013704      92        469630.4
##   4: E01013602      58        217060.3
##   5: E01013672      50        432440.0
##  ---
## 337: E01034013      13        219423.1
## 338: E01013638      17        199823.5
## 339: E01013748      11        205090.9
## 340: E01025787       3        336666.7
## 341: E01025687      14        340357.1
```

Merge before and after summaries onto the map

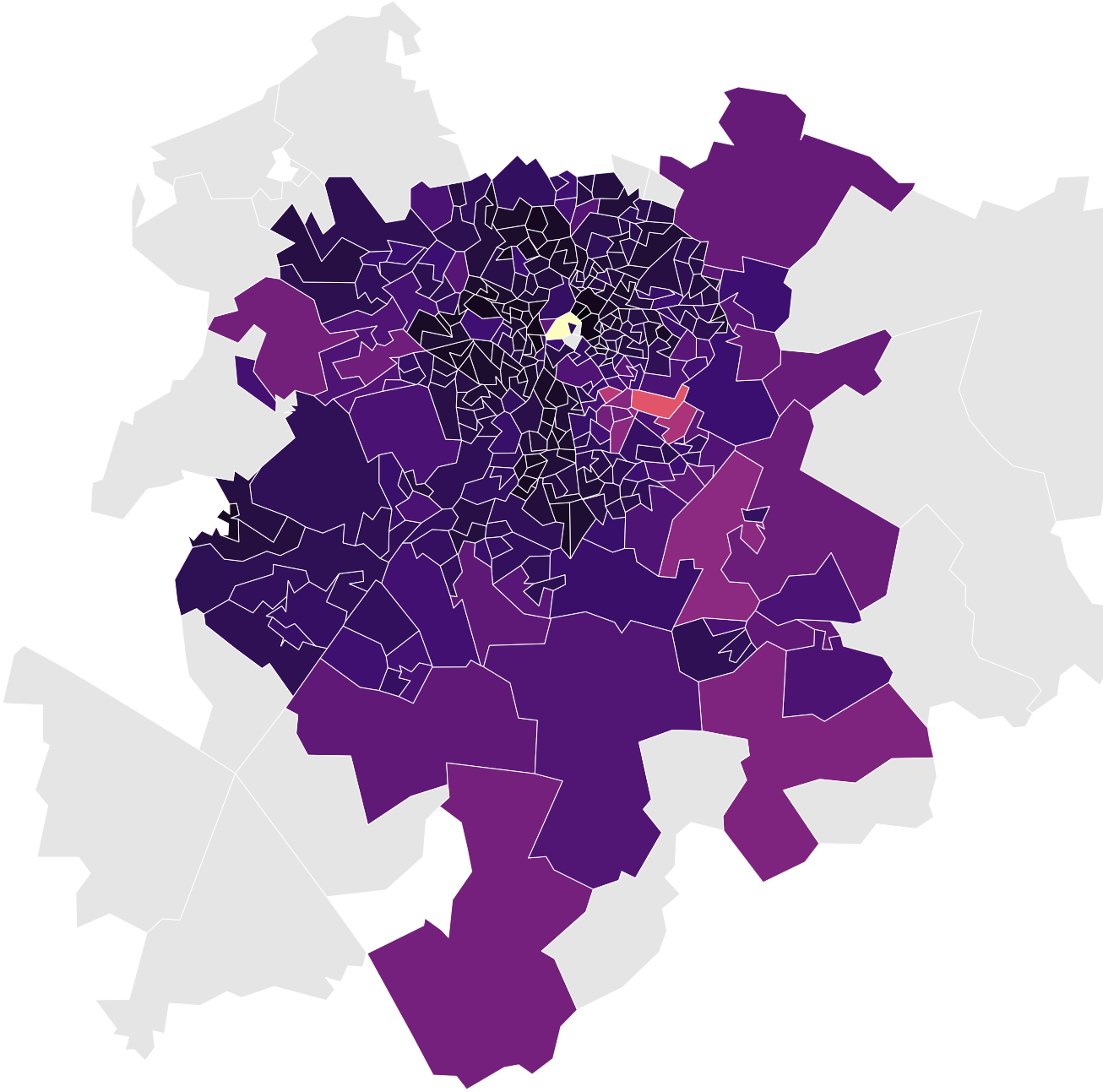
```r
map_before <- merge(lsoa_sf, before_prices, by = "lsoa21cd", all.x = TRUE)
map_after <- merge(lsoa_sf, after_prices, by = "lsoa21cd", all.x = TRUE)
```

Heatmap 1: Average house price by LSOA before the event

```r
ggplot(map_before) +
  geom_sf(aes(fill = avg_price_before), color = "white", linewidth = 0.1) +
  scale_fill_viridis_c(
    option = "magma",
    na.value = "grey90",
    labels = label_dollar(prefix = "£")
  ) +
  labs(
    title = "Average House Price by LSOA (Before Event)",
    subtitle = paste0(
      "Event date cutoff: ", event_date,
      " | LSOAs with fewer than ", min_sales_per_lsoa, " sales are omitted"
    ),
    fill = "Average price"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold"),
    axis.title = element_blank(),
    axis.text = element_blank(),
    panel.grid = element_blank()
  )
```

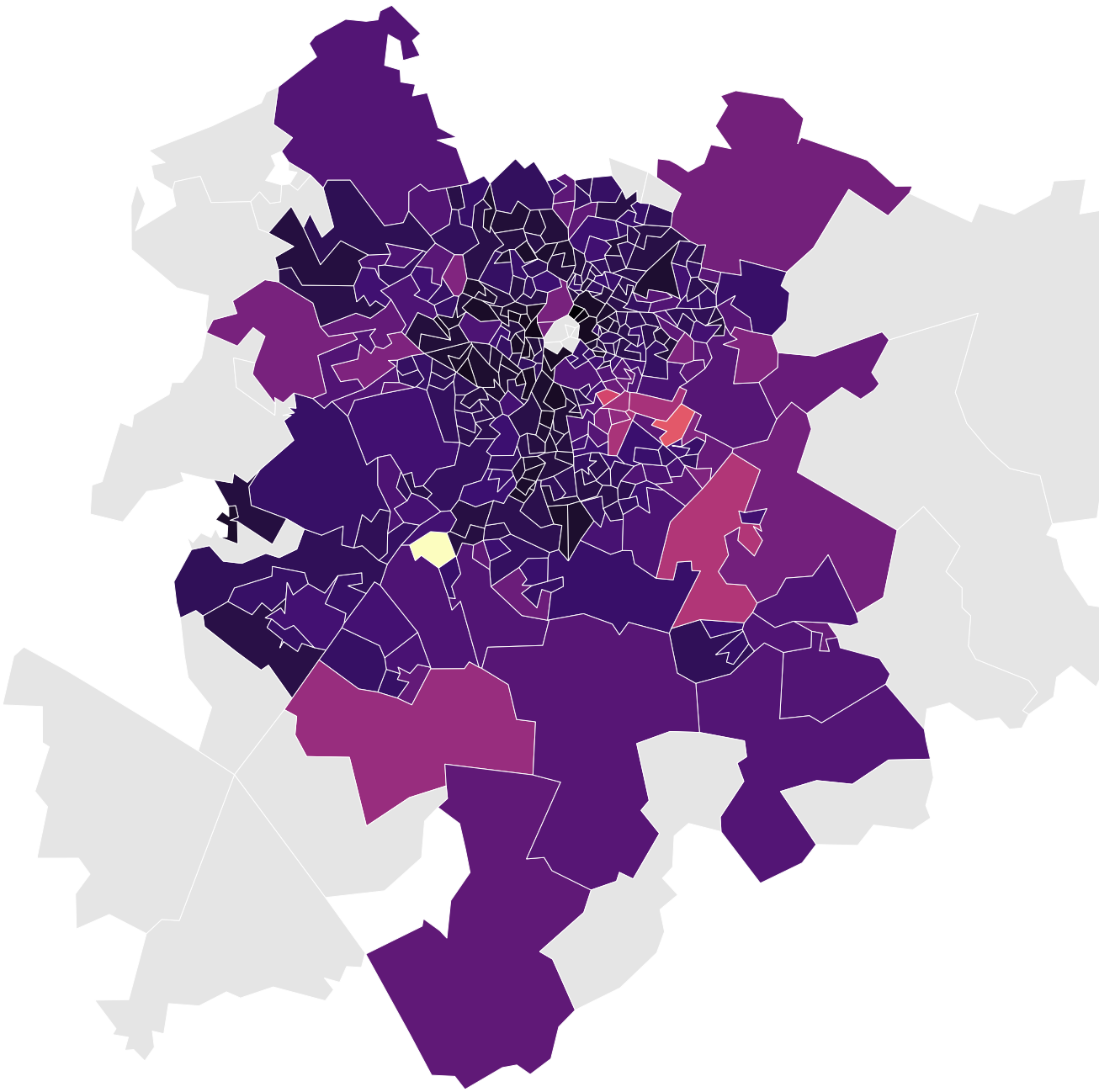
Average House Price by LSOA (Before Event)

Event date cutoff: 2023−05−01 | LSOAs with fewer than 3 sales are omitted



Heatmap 2: Average house price by LSOA after the event

```r
ggplot(map_after) +
  geom_sf(aes(fill = avg_price_after), color = "white", linewidth = 0.1) +
  scale_fill_viridis_c(
    option = "magma",
    na.value = "grey90",
    labels = label_dollar(prefix = "£")
  ) +
  labs(
    title = "Average House Price by LSOA (After Event)",
    subtitle = paste0(
      "Event date cutoff: ", event_date,
      " | LSOAs with fewer than ", min_sales_per_lsoa, " sales are omitted"
    ),
    fill = "Average price"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold"),
    axis.title = element_blank(),
    axis.text = element_blank(),
    panel.grid = element_blank()
  )
```



Build relative house price inflation

The inflation measure used here is:

$$\text{Inflation (\%)} = \left(\frac{\text{Average Price After}}{\text{Average Price Before}} - 1 \right) \times 100$$

```
inflation_dt <- merge(
  before_prices[, .(lsoa21cd, n_before, avg_price_before)],
  after_prices[, .(lsoa21cd, n_after, avg_price_after)],
  by = "lsoa21cd",
  all = FALSE
)

inflation_dt <- inflation_dt[avg_price_before > 0]

inflation_dt[, inflation_pct := ((avg_price_after / avg_price_before) - 1) * 100]

inflation_dt
```

```
## Key: <lsoa21cd>
##       lsoa21cd n_before avg_price_before n_after avg_price_after inflation_pct
##         <char>    <int>            <num>   <int>           <num>         <num>
##   1: E01013600      240         121795.3      34        213941.2     75.656330
##   2: E01013601      241         135192.2      51        238776.5     76.620050
##   3: E01013602      396         138734.5      58        217060.3     56.457378
##   4: E01013603      192         135495.8      37        195925.5     44.598887
##   5: E01013604      231         126425.8      48        196041.7     55.064656
##  ---
## 336: E01034025      481         227216.5      68        348125.0     53.212894
## 337: E01035084      745         266762.4     103        330987.0     24.075600
## 338: E01035085      457         305700.7      72        333972.2      9.248089
## 339: E01035092      604         242252.1      81        284049.4     17.253641
## 340: E01035093      471         284769.3     105        345847.6     21.448374
```

Merge inflation onto the map

```
map_inflation <- merge(lsoa_sf, inflation_dt, by = "lsoa21cd", all.x = TRUE)
```

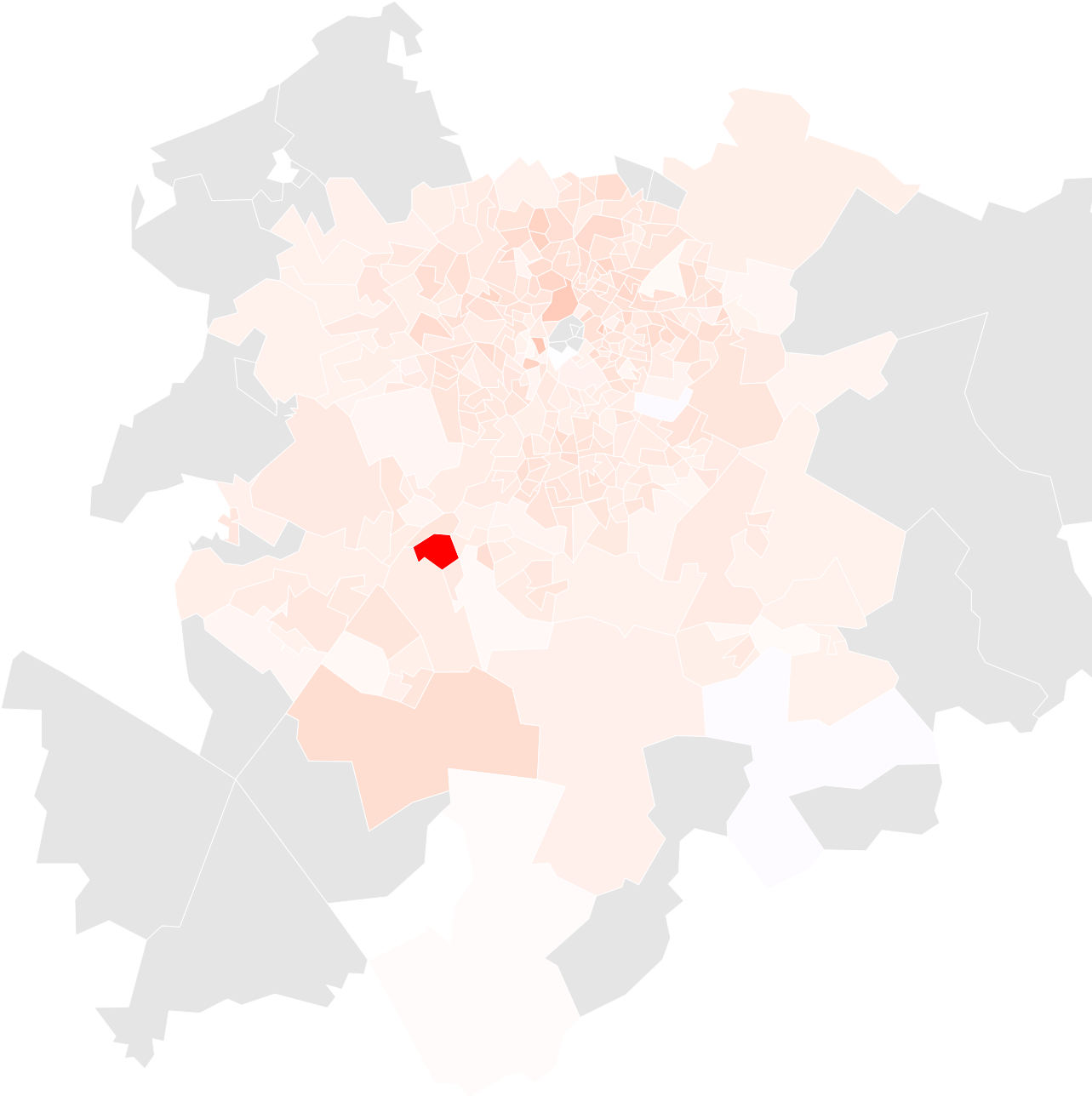
Heatmap 3: Relative house price inflation by LSOA

```
ggplot(map_inflation) +
  geom_sf(aes(fill = inflation_pct), color = "white", linewidth = 0.1) +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 0,
    na.value = "grey90",
    labels = function(x) paste0(round(x, 1), "%")
  ) +
  labs(
```

```r
    title = "Relative House Price Inflation by LSOA",
    subtitle = paste0(
      "Inflation = ((Average Price After / Average Price Before) - 1) × 100",
      " | Only LSOAs with at least ", min_sales_per_lsoa,
      " sales in both periods are shown"
    ),
    fill = "Inflation %"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold"),
    axis.title = element_blank(),
    axis.text = element_blank(),
    panel.grid = element_blank()
  )
```

Relative House Price Inflation by LSOA

Inflation = ((Average Price After / Average Price Before) − 1) × 100 | Only LSOAs with at least 3 sales in



Optional tables for reporting

```
sales_by_period <- houses[, .(
  rows = .N,
  unique_lsoas = uniqueN(lsoa21cd)
), by = period]

sales_by_period
```

```
##    period   rows unique_lsoas
##    <char>  <int>        <int>
## 1: Before 114184          346
## 2:  After  21142          346
```

```
map_counts <- data.table(
  map = c(
    "Average price before",
    "Average price after",
    "Relative inflation"
  ),
  lsoas_with_values = c(
    sum(!is.na(map_before$avg_price_before)),
    sum(!is.na(map_after$avg_price_after)),
    sum(!is.na(map_inflation$inflation_pct))
  )
)

map_counts
```

```
##                      map lsoas_with_values
##                   <char>             <int>
## 1: Average price before               345
## 2:  Average price after               341
## 3:   Relative inflation               340
```